

Prompt Engineering for Beginners

How to Write Better Prompts and Get Better AI Responses

What is a Prompt?

A **prompt** is the question or instruction you give to an AI system. Think of it as a conversation starter with a very knowledgeable assistant.

Example:

-  Vague: "Help me"
 -  Good: "Help me write a friendly email to my child's teacher about extending a homework deadline"
-

Why Good Prompts Matter

Better prompts = Better responses!

Bad Prompt: "Food ideas" **AI Response:** *Generic list of recipes*

Good Prompt: "Suggest 3 healthy dinner ideas for a family of 4, including one vegetarian child, budget £20, taking 30 minutes or less" **AI Response:** *Specific, useful suggestions matching your exact needs*

The 5 Elements of a Great Prompt

1. Context - Give Background

Who you are, what situation you're in

Example: "I'm a parent of a 10-year-old who struggles with focus..."

2. Task - Be Specific

What exactly do you want

Example: "...help me create a homework routine that works for children with ADHD..."

3. Format - How You Want It

List, paragraph, step-by-step, table, etc.

Example: "...as a simple visual checklist..."

4. Constraints - Set Boundaries

Length, tone, level, time, budget

Example: "...that takes no more than 45 minutes total..."

5. Examples - Show What You Mean

Provide examples of what you want

Example: "...similar to the one attached, but with more specific timings"

The Before & After: Real Examples

Example 1: Meal Planning

 **Weak Prompt:** "Meal ideas"

 **Strong Prompt:** "Create a weekly dinner menu for a family of 4 (2 adults, 2 children ages 6 and 9). One child is vegetarian. Budget: £75. Must include nutritional balance and

use common ingredients from UK supermarkets. Provide a shopping list organized by supermarket section."

Why it's better:

- Context: family size and ages
 - Specific requirements: vegetarian, budget
 - Constraints: UK ingredients
 - Format requested: organized shopping list
-

Example 2: Email Writing

✗ Weak Prompt: "Write an email"

✓ Strong Prompt: "Help me write a polite but firm email to my internet provider. Context: We've had 3 outages this month, I've reported each one. I want a partial refund and assurance this will be fixed. Tone: professional but assertive. Length: 3 short paragraphs maximum."

Why it's better:

- Context: specific problem
 - Task: clear outcome wanted
 - Tone specified
 - Length constraint
-

Example 3: Homework Help

✗ Weak Prompt: "Explain photosynthesis"

✓ Strong Prompt: "Explain photosynthesis to a 9-year-old in simple terms. Use a real-world analogy they'd understand. Include the main steps but avoid complex scientific terms. Follow with 3 questions I can ask my child to check they understood."

Why it's better:

- Audience specified: 9-year-old
- Method: analogy

- Constraints: simple language
 - Extra request: check questions
-

Prompt Formulas That Work

Formula 1: The Clear Request

```
"I need to [TASK] for [AUDIENCE].  
Please [SPECIFIC ACTION]  
in a [TONE/STYLE] way.  
Keep it [CONSTRAINT]."
```

Example: "I need to explain budgeting to my teenager. Please create a simple guide in a casual, non-preachy way. Keep it under 500 words."

Formula 2: The Problem-Solver

```
"I'm facing [PROBLEM] in [SITUATION].  
My goal is [DESIRED OUTCOME].  
Suggest [NUMBER] practical solutions  
considering [CONSTRAINTS]."
```

Example: "I'm facing difficulty getting my 7-year-old to bed on time during school nights. My goal is a consistent 8pm bedtime. Suggest 5 practical solutions considering we have limited evening time and he has lots of energy."

Formula 3: The Learning Helper

```
"Explain [CONCEPT] to someone who [KNOWLEDGE LEVEL].  
Use [TEACHING METHOD].  
Include [ADDITIONAL REQUESTS]."
```

Example: "Explain fractions to someone who understands basic division. Use visual examples with pizza or pie. Include 3 practice problems that gradually increase in difficulty."

Formula 4: The Creative Brief

```
"Create [DELIVERABLE] for [PURPOSE].  
Style: [TONE/STYLE]  
Include: [SPECIFIC ELEMENTS]  
Avoid: [THINGS TO EXCLUDE]"
```

Example: "Create a birthday party plan for my 8-year-old's Harry Potter themed party. Style: magical and fun but achievable. Include: games, decorations we can make at home, menu, timeline. Avoid: expensive store-bought items or things requiring special skills."

Power Words That Improve Prompts

For Better Explanations:

- "Explain like I'm 5/10/a beginner..."
- "Break this down into simple steps..."
- "Use an everyday analogy..."
- "What's a real-world example of..."

For Better Structure:

- "Organize this as..."
- "Create a step-by-step guide..."
- "Format this as a table/list/outline..."
- "Structure this from most to least important..."

For Better Tone:

- "Write in a friendly/professional/casual tone..."

- "Make this more encouraging/direct/gentle..."
- "Keep the language simple/formal/conversational..."

For Better Depth:

- "Give me 3 specific examples..."
 - "Provide detailed reasoning..."
 - "Include pros and cons..."
 - "What are the main points in 3 bullet points..."
-

The "Ask Me Questions" Trick

If you're not sure how to write a good prompt, try this:

Your Prompt: "I need help planning a family holiday but I'm not sure what details you need. Please ask me questions to help create the perfect itinerary."

AI will respond: "I'd be happy to help! Let me ask you some questions:

1. How many people? Ages?
2. Preferred destination type?
3. Budget range?
4. Duration?
5. Interests and activities? ..."

Then you answer, and AI creates exactly what you need!

Common Prompt Mistakes

✗ Mistake 1: Too Vague

"Help me with dinner"

Fix: Be specific! "Suggest a quick vegetarian dinner recipe for 2 that uses ingredients I likely have: pasta, tomatoes, cheese, onions, garlic"

✗ Mistake 2: Too Many Tasks at Once

"Help me plan my week, organize my finances, create a meal plan, and write emails"

Fix: One task at a time! Start with: "Help me create a weekly meal plan. Once we finish that, I'll ask about the other tasks."

✗ Mistake 3: Assuming AI Knows Context

"How do I fix it?"

Fix: Provide context! "My dishwasher isn't draining properly. Water sits at the bottom after a cycle. What are common causes and how can I fix each one?"

✗ Mistake 4: Not Specifying Tone/Style

"Write a complaint letter"

Fix: Specify tone! "Write a firm but polite complaint letter about faulty goods, requesting a full refund. Professional tone, 3 paragraphs maximum."

✗ Mistake 5: Forgetting Your Audience

"Explain quantum physics"

Fix: Specify level! "Explain quantum physics to a curious 12-year-old who loves science. Use simple analogies and avoid complex maths."

Iterating: Making Responses Better

First response not quite right? **Refine it!**

Round 1:

You: "Give me birthday party ideas for a 6-year-old"

AI: *Gives generic list*

Round 2:

You: "Make these more specific for a dinosaur-loving child. Include activities that work indoors since it's a December party. Budget: £100."

AI: *Gives better, tailored ideas*

Round 3:

You: "Perfect! Can you now create a timeline for the 2-hour party, including setup time?"

AI: *Provides detailed schedule*

Key Point: It's a conversation! Ask follow-ups to refine.

Prompt Templates for Families

Template 1: Homework Help

"My [AGE]-year-old is learning about [TOPIC] in school.
Explain [SPECIFIC CONCEPT] in simple terms
using [TYPE OF EXAMPLE].
Then give me 3 questions to check understanding."

Template 2: Meal Planning

"Create a [NUMBER] day meal plan for [FAMILY SIZE].
Dietary needs: [REQUIREMENTS].
Budget: [AMOUNT].
Include [SPECIFIC REQUESTS: easy recipes/shopping list/prep time].
Avoid [DISLIKES/ALLERGIES]."

Template 3: Activity Planning

```
"Suggest [NUMBER] [INDOOR/OUTDOOR] activities
for [AGE RANGE].
Weather: [CONDITION].
Budget: [AMOUNT].
We have [AVAILABLE MATERIALS].
Time available: [DURATION]."
```

Template 4: Email Writing

```
"Help me write a [TONE: professional/friendly/firm] email to [REC
Context: [SITUATION].
I want to [GOAL].
Keep it [LENGTH].
[SPECIAL REQUESTS]."
```

Template 5: Problem Solving

```
"I'm struggling with [PROBLEM].
Context: [RELEVANT DETAILS].
I've already tried [PREVIOUS ATTEMPTS].
Suggest [NUMBER] alternative solutions
that consider [CONSTRAINTS]."
```

The Follow-Up Questions to Always Ask

After getting an AI response, improve it with:

1. "Can you make this simpler?"

- Reduces complexity

2. "Can you give me a specific example?"

- Makes it more concrete

3. "What are the pros and cons of this approach?"

- Adds critical thinking

4. "Are there any alternatives?"

- Gives you options

5. "Can you format this as [LIST/TABLE/STEPS]?"

- Makes it easier to use

Privacy-Safe Prompt Rewrites

Never share personal details! Rewrite prompts to be safe:

Unsafe:

"My daughter Emma, age 8, at Willow Primary School in Leeds is being bullied by a classmate named Sophie"

Safe:

"My 8-year-old is experiencing bullying from a classmate. How should I approach this with the school?"

Unsafe:

"I live at 24 Oak Road, Manchester M13 4PQ. What's the best internet provider?"

Safe:

"I'm in the Greater Manchester area. What internet providers have good service here?"

Testing Your Prompts: The Checklist

Before sending your prompt, check:

☐ **Is it specific enough?** (Who, what, when, where, why, how) ☐ **Did I provide context?** (Background information) ☐ **Did I specify format?** (How I want the response) ☐ **Did I set constraints?** (Length, tone, level) ☐ **Did I remove personal information?** (Privacy check) ☐ **Is it one clear task?** (Not multiple requests)

Practice Exercises

Try improving these prompts:

Exercise 1:

✗ Weak: "Travel ideas"

Your improved version:

Suggested answer: "Suggest 3 family-friendly UK holiday destinations for summer. Family: 2 adults, kids aged 5 and 8. Budget: £1500 including accommodation. Interested in outdoor activities and historical sites. 4-5 days duration."

Exercise 2:

✗ Weak: "Help with homework"

Your improved version:

Suggested answer: "My 10-year-old needs help understanding long division. Explain the process step-by-step using a simple example. Then provide 3 practice problems with answers I can use to check their work."

Exercise 3:

✗ Weak: "Party ideas"

Your improved version:

Suggested answer: "Plan a 13th birthday party for my sports-loving teen. 15 guests, mix of boys and girls. Budget: £200. Need: activity ideas that aren't childish but not adult either, food suggestions, and a 3-hour timeline. Indoor venue at our home."

Advanced Tips

Tip 1: Role Prompting

Ask AI to take on a specific role:

"Act as a primary school teacher. Explain [concept] to a 7-year-old..."

"Act as a financial advisor. Review this family budget and suggest improvements..."

Tip 2: Chain of Thought

Ask AI to show its reasoning:

"Explain your thinking step-by-step as you solve this problem..."

Tip 3: Constraint Creativity

Constraints often make responses better:

"In exactly 100 words..." "Using only common household items..." "Without spending any money..."

Quick Reference: Prompt Starters

For Learning:

- "Explain [topic] to a [age]-year-old..."
- "What's a simple way to understand..."
- "Break down [concept] into steps..."

For Planning:

- "Create a [timeframe] plan for [goal]..."
- "Organize [event] considering [constraints]..."
- "What's a realistic schedule for..."

For Writing:

- "Help me write a [type] to [recipient] about [topic]..."
- "Improve this [text] to make it more [quality]..."
- "Draft a [document] that includes [elements]..."

For Problems:

- "I'm struggling with [issue]. Suggest solutions that..."
 - "What are different approaches to [problem]..."
 - "How can I [goal] while avoiding [obstacle]..."
-

Remember

🎯 **Be Specific** - The more details, the better 🎯 **Provide Context** - Help AI understand your situation 🔄 **Iterate** - Refine responses with follow-ups 🛡️ **Stay Safe** - Never share

personal information 🎯 **Think Critically** - AI can make mistakes

Most Important: Practice makes perfect! The more you practice writing prompts, the better your results will be.

Resources

Practice More:

- Alongside AI Playground: alongside-ai-basics.netlify.app/playground
- Prompt Library: alongside-ai-basics.netlify.app/prompts

Learn More:

- The AI Ark: theaiark.org
 - Full course modules: alongside-ai-basics.netlify.app
-

This guide is provided by The AI Ark Free AI Education for Families theaiark.org